

container-based implementation of your desired database and Web framework can be consumed directly rather than you having to fork them both and rework them to be compatible.

BOSH – from the Cloud Foundry

BOSH is a cloud-agnostic open source tool for release engineering, deployment and life cycle management of complex distributed systems. Most distributions and managed Cloud Foundry environments use BOSH to holistically manage the environment so that you can focus on coding and delivering business value. Due to the flexibility and power of BOSH, Google and Pivotal made it the heart of the Kubo project (Web-scale release engineering for Kubernetes).

BOSH works like a distributed system, where collections of individual software components, running on different virtual machines, work together as a larger system.

The main components of BOSH are listed below.

Stem cell: This is similar to a golden image of an OS used to create virtual machines. It isolates the base operating system from the other software packages bundled in a deployment.

Release: This is a layer put on top of a stem cell that describes what software should be deployed and how it should be configured. It contains configuration properties and templates, startup scripts, source code, binary artifacts, etc.

Deployment manifest: BOSH uses this YAML manifest file to deploy to the targeted infrastructure, monitor the health of the virtual machines or containers, and repair them when necessary.

Cloud Provider Interface: CPI is an API that BOSH uses to interact with infrastructure to create and manage stem cells, VMs and disks.

BOSH is being adopted in other environments to package and manage all kinds of software. BOSH's strength is that it manages both Day One and Day Two operational tasks, such as getting software configured and upgrading it to new versions, testing the upgrade of an entire system

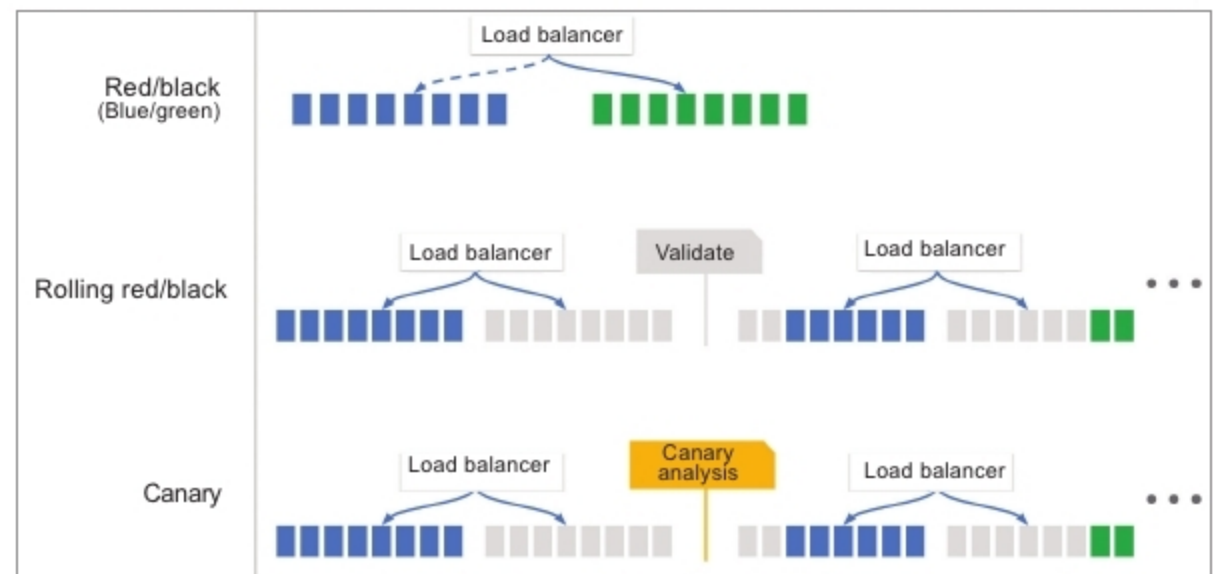


Figure 6: The Spinnaker deployment strategies

from one version to another, resizing host machines, and handling security updates—without disrupting users—consistently and reliably.

Spinnaker

Spinnaker is an open source, multi-cloud continuous delivery platform for releasing software changes with high velocity and confidence. It combines a powerful and flexible pipeline management system with integrations from major cloud provider services like AWS EC2, Kubernetes, Google Compute Engine, Google Kubernetes Engine, Google App Engine, Microsoft Azure, OpenStack, Cloud Foundry, and Oracle Cloud Infrastructure.

It automates the releases by creating deployment pipelines, which run integration and system tests, spin up and down server groups, and monitor your rollouts. It triggers pipelines via Git events, Jenkins, Travis CI, Docker, CRON, or other Spinnaker pipelines.

Spinnaker creates and deploys immutable images for faster rollouts, easier rollbacks and the elimination of hard-to-debug configuration drift issues. It leverages an immutable infrastructure in the cloud with built-in deployment strategies such as red/black and canary deployments.

Spinnaker's application management feature helps you to manage the cloud resources. Spinnaker operates as a collection of services—sometimes referred to as applications or microservices. Applications, clusters

and server groups are the key concepts Spinnaker uses to describe the services. Load balancers and firewalls describe how your services are exposed to users.

The Spinnaker pipeline is the key deployment management construct and it consists of a sequence of actions, known as stages. You can pass parameters from stage to stage along the pipeline (as shown in Figure 5). You can start a pipeline either manually or automatically trigger it by an event, such as the completion of a job, a new image appearing in your registry, a CRON schedule or a stage in another pipeline. You can configure the pipeline to give notifications by email, Slack, or SMS, to interested parties at various points during pipeline execution.

Spinnaker treats cloud-native deployment strategies as first-class constructs, handling the underlying orchestration such as verifying health checks, disabling old server groups and enabling new server groups. Spinnaker supports the red/black (a.k.a. blue/green) strategy, with rolling red/black and canary strategies in active development. Do refer to Figure 6 for the details. **END** 

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